

ReviewBot Assessment Report

The Face-Adjacency Graph of the Truncated Octahedron

Luke Martin — Draft 5, October 2026

Evidence-based, bounded memetic review of a mathematical manuscript
Thirteen persistent logical ReviewBot agents; one computational reviewer runtime

8 October 2026

Prepared in the reader-facing assessment format of the supplied August 2026 ReviewBot example.
ReviewBot 3.0 discussion policies were used for the bounded review-search pilot;
this is *not* a claim of a calibrated, independent multi-LLM system.

Reviewer’s note on scope. The assessment covers the supplied 12-page Draft 5 PDF and recovered, pinned previous verification archives, not an unexamined subsequent manuscript revision. The previously recovered mathematical checks and audit findings are distinguished below from the **14 fresh tasks actually executed** in the new algorithmic review run. The run has a genuine thirteen-position Current/Pocket/Pareto trace, with five substantive generations; it has *no independent LLM worker population, no Lean compilation, and no calibrated search performance*. Mathematical errors, incomplete evidence, and publication-readiness issues are reported separately.

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1 Executive Summary

- **Manuscript type:** A (pure mathematics with computational verification). The conditional metrics M15 (gauge structure) and M18 (algebraic structure) apply. M16 and M17 are inactive for this standalone mathematical note.
- **Overall manuscript assessment:** approximately **8.9/10**. This is the recovered, adapted rubric assessment ($Q = 8.871$) retained after the new checks; it is *not* a measurement of ReviewBot fitness or confidence in correctness.
- **Recommendation: Accept with minor revisions, subject to venue fit, and requiring a verification/reproducibility clean-up before clearance.** Mathematical validity gate: no confirmed fatal or major mathematical error in the reviewed material, but no claim of a complete proof-assistant certificate.
- **Errors/issues:** 0 confirmed critical mathematical errors, 0 confirmed major mathematical errors; **6 grouped minor manuscript/release corrections** (five inherited, one added under the new proof-disclosure requirement), and 3 optional exposition groups.
- **Fresh review-search result: 5 substantive refinement generations;** 14/14 frozen checking tasks executed and passed; 30 Current-to-Pocket improvements, 16 bottom-up Pocket promotions. Stop reason: FROZEN_TASK_SCOPE_EXHAUSTED; neither stability convergence nor proof completeness is asserted.

Top three strengths.

1. The ordinary Laplacian factorization and O_h representation blocks are transparent: every eigenvalue and multiplicity can be reconstructed from the square/hexagon orbit structure.
2. At quarter flux the explicit incidence Gram identity, six two-dimensional invariant blocks, a restricted intertwiner, and a magnetic rotation representation explain rather than merely fit the special spectrum.
3. The uniform-flux sector classification is a concise exact third-moment argument: $\text{tr}(\Delta_q^3) = 3264 - 144 \cos(\pi q/12)$, proving precisely thirteen distinct spectra.

Top three limitations.

1. An otherwise sound computational release still has a loose numerical tolerance contract, a rounded determinant described as an exact certificate, and a mismatched ordered lift label.
2. Publication provenance and installer dependency statements need precise correction; missing machine-proof disclosure must be supplied under the agreed newer ReviewBot standard.
3. The note has specialist value but limited generality; this review does not establish historical priority, physical validity of UFFT, or exact individual spectra at all charges.

2 Category-Level Breakdown

Category	Effective weight	Category score	Weighted contribution
Correctness (M1–M4)	45%	9.00	4.05
Completeness (M5–M7)	22.5%	9.02	2.03
Clarity (M8–M10)	18%	8.87	1.60
Significance (M11–M14)	4.5%	6.22	0.28
Specialized M15: Gauge	7%	9.00	0.63
Specialized M18: Algebra	3%	9.50	0.29
Total	100%		8.87/10

The pure-mathematics category weights (50, 25, 20, 5)% are renormalized to 90% of the total after 7% gauge and 3% algebra specialization. Internal category averages use the original published within-category metric proportions. Displayed contributions are rounded. No optimization operator attempts to increase this paper score.

3 Detailed Metric Assessment

3.1 Category 1: Mathematical Correctness

Metric	Score	Justification
M1 Preci- sion	9.0	Theorems and propositions delimit the graph, connected spherical cellular embedding, $U(1)$ phases, integer uniform sectors, and matrix domains. No physical UFFT identification is relied on. The restricted sector of the intertwiner is stated.
M2 Proofs	9.0	The orbit blocks, degree-of-freedom count, planar-tree gauge classification, third-moment distinction and Gram reduction are worked out in the manuscript. Prior exact scripts corroborate finite identities but do not constitute Lean-checked proofs.
M3 Calcu- lation	9.0	Fresh construction from the 24 original polyhedron vertices, both characteristic polynomials, 101,154,816 spanning trees, two special-charge spectra, Gram identity, all-charge moments, and gauge covariance passed. Five code/release corrections still matter.
M4 Con- sistency	9.0	The four-dimensional eigenvalue-7 space is correctly split $A_{1g} \oplus T_{2g}$; K is not claimed to annihilate its A_{1g} line. Gauge and rotational labels distinguish physical invariants from representative choices. The compensating-plaquette ordering needs one correction.

3.2 Category 2: Mathematical Completeness

Metric	Score	Justification
M5 Defi- nitions	9.0	The graph, P_s, P_h, A_{sh} , degrees, support weights, magnetic connection and flux sector are defined. A displayed gauge-transformation law and explicit $A_{hs} = A_{sh}^\top$ at first appearance would help.
M6 Assump- tions	8.8	Sphere topology, face orientation and representation scope are explicit. The executable tolerance, determinant, and checker coverage claims should match their precise evidence.
M7 Scope	9.3	The note avoids asserting a general tiling-selection principle, physical space ontology, continuous eigenvalue branches, or exact formulas at all charges. The five Fedorov types are a descriptive comparison, not a physical deduction.

3.3 Category 3: Mathematical Clarity

Metric	Score	Justification
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M8 Notation	8.8	Orbit labels, integer charge q , eigenvalues r_1, r_2 and parity notation are mostly stable. The face-0 compensating term is not positioned consistently in prose and certificate. The alleged missing prism overbar was a prior retracted extraction error; it is not counted as a defect.
M9 Accessibility	9.0	The raw/orthonormal orbit blocks, cube Hamming-distance argument and invariant-plane reduction make a specialized result readable. A short roadmap would improve the longest magnetic argument.
M10 Organization	8.8	The graph, spectrum, support, antisymmetric operator, magnetic sectors, and five-cell comparison occur in natural order; calculation and release claims in Section 8 need careful alignment.

3.4 Category 4: Mathematical Significance

Metric	Score	Justification
M11 Originality	6.0	The explicit special-charge blocks, intertwiner and classification look like useful example-specific contributions. An exhaustive historical priority search was not undertaken.
M12 Generality	4.0	The analysis applies in detail to one face-adjacency graph, with exact magnetic eigenvalues in selected charge sectors. A larger theorem about arbitrary networks or space microstructures is not claimed.
M13 Literature	8.7	Relevant graph-spectral, spherical-magnetic and gain-graph literature is acknowledged. Source relevance was checked in the inherited referee review; new exhaustive bibliographic research was outside this run.
M14 Impact	5.5	Potentially a clean benchmark and teaching case for magnetic graph Laplacians, not a resolution of an outstanding physical question.

3.5 Specialized Metrics

Metric	Score	Justification
M15 Gauge	9.0	Vertex phases, spherical holonomy constraints, integer lifts and gauge-invariant spectra are differentiated. Our alternative tree construction gives an independent covariance check at seven charges.
M16 Break-through	N/A	No new paradigmatic claim is made in this standalone manuscript.
M17 Synthesis	N/A	The note does not claim to unify different physical theories.
M18 Algebra	9.5	Exact integer and Gaussian-integer characteristic polynomials, B^*B , BB^* , finite-group actions and the moment identity are appropriate finite algebraic objects. They do not require a Lie-algebra protocol.

4 Error Report and Independent Verification

4.1 Re-derived correct results (new review run)

V1. Polyhedral incidence (T01–T02). Starting from the 24 distinct permutations of $(0, \pm 1, \pm 2)$, build six square and eight hexagon supporting-face vertex sets. Two faces meet along a polyhedron edge when their vertex sets intersect in two points. The resulting graph has 14

face vertices, 36 graph edges, degree sequence $4^6 6^8$ and 24 triangular plaquettes. Its exact Laplacian characteristic polynomial is

$$x(x-9)(x-7)^4(x-4)^2(x^2-9x+16)^3,$$

with the corresponding adjacency characteristic polynomial of Eq. (1). This independently reproduces the geometry without importing the released graph-edge list.

V2. Matrix–Tree cross-check invariant (T03). Exact deletion of one row and column from the 14×14 Laplacian gives

$$\tau(\Gamma) = 101,154,816 = \frac{9 \cdot 7^4 \cdot 4^2 \cdot 16^3}{14}.$$

The second expression follows independently from the nonzero eigenvalues. The agreement is an additional check; it is not a new claim in Draft 5.

V3. Additional exact spectral moment (T04). With $z = e^{i\pi q/12}$, integer edge-exponent lifts and complete length-four closed-walk enumeration yield in the finite group ring $\mathbb{Z}[z]/(z^{24}-1)$:

$$\text{tr } \Delta_q^4 = 25080 - 1536(z + z^{-1}) + 144(z^2 + z^{-2}) + 24(z^4 + z^{-4}).$$

Equivalently,

$$\text{tr } \Delta_q^4 = 25080 - 3072 \cos(\pi q/12) + 288 \cos(\pi q/6) + 48 \cos(\pi q/3).$$

The independent length-three enumeration recovers the manuscript’s $3264 - 144 \cos(\pi q/12)$. This is *new auditor-generated auxiliary evidence*, not a novelty or authorship assertion about the manuscript.

V4. Special magnetic sectors (T06–T08). Exact symbolic verification gives $B^*B = (9I - C^2)/2$ and $BB^* = 4I_6$. A new spanning-tree gauge, chosen differently from the archived certificate, reproduces the complete characteristic polynomials at $q = 6$ and $q = 12$. The $q = 12$ signless-Laplacian calculation also passes directly.

V5. Whole sweep and gauge invariance (T05, T09). The new tree-based gauge passes all 25 charge sectors with maximum first-to-third-moment numerical residual 2.73×10^{-12} . Two different spanning trees produce connections related by an explicit vertex gauge; their matrices were compared at seven charges.

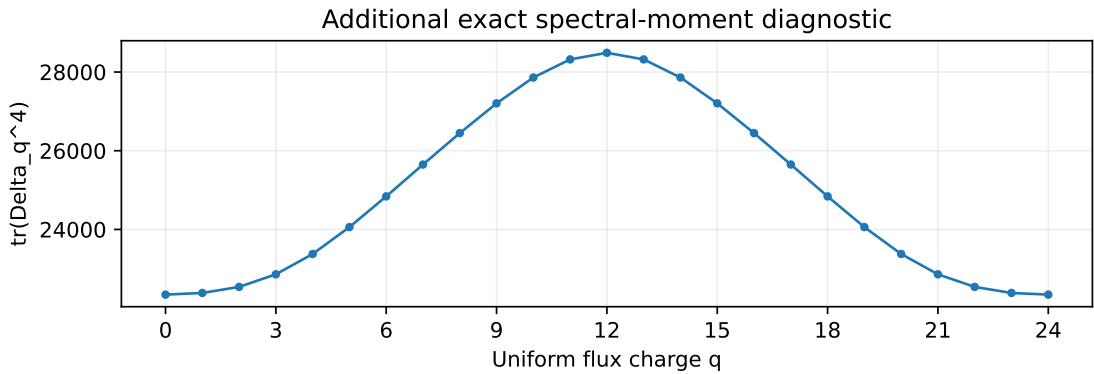


Figure 1. Exact fourth-moment values over the 25 discrete charge labels, computed from the new Laurent-polynomial coefficient table. Lines guide the eye only.

4.2 Issues requiring correction

R1 — Numerical tolerance contract (minor; confirmed). Section 8 and the released plotting script describe absolute 10^{-9} checks, but five calls to `np.allclose` in the archived plotting

implementation omit `rtol=0`. Our fresh negative test shows that 8.00005 versus 8 can pass `atol=1e-9` with default relative tolerance but fails when `rtol=0`. *Fix*: specify `rtol=0`, explicitly set the complex-holonomy tolerance, and rerun the release plotter (the earlier strict computational check passed).

R2 — Exactness of the unimodularity certificate (minor; confirmed). The archived plotter computes and rounds a floating determinant to support a unimodularity assertion. A rounded float is not an exact integer certificate. Our exact determinant calculation gives +1 for the released 23×23 minor, and another +1 for the new spanning-tree minor. *Fix*: use exact integer arithmetic or narrow the description to a numerical check. The manuscript’s exact face residues and lift relations were already sufficient for the claimed holonomies; this is *not* a missing-holonomy-proof allegation.

R3 — Ordered lift vector (minor; confirmed). Section 8 describes the target as $(1, \dots, 1, -23)$, whereas the released face-0 ordering uses $(-23, 1, \dots, 1)$. Both are legitimate after consistent reindexing, but the existing certificate should be described without silent permutation. *Fix*: use $(-23, 1, \dots, 1)$ and identify face 0.

R4 — Historical audit provenance (minor; confirmed documentary mismatch). Proposition 6.2, Section 6.1 and Appendix A retain the phrase “independent audit” for earlier rounds, while the acknowledgements credit a memetic ReviewBot system. The recovered evidence documents AI-assisted mathematical review and reproducible programs; it does not, by itself, demonstrate that the four historical rounds executed a complete memetic optimizer. *Fix*: describe them as successive review rounds, qualify independence at the code/check level where supported, and distinguish the present bounded pilot from any past implementation. Do not retroactively relabel earlier reviews as a new memetic run.

R5 — Release dependencies (minor; confirmed). The archived `requirements.txt` contains NumPy, SymPy and NetworkX but not `matplotlib`, while the figure module imports it and the paper lists it. *Fix*: add the dependency and give one tested installation/runner route; keep NetworkX optional where appropriate.

R6 — Proof-checking disclosure (minor; newly requested under the agreed policy). The manuscript does not declare whether any mathematical statements have been validated by Lean or another checker. The recovered evidence gives none. *Fix*: add a visible proof-verification statement and a theorem/claim-level dependency table saying *not machine checked* where that is the case. This is a transparency obligation, *not* evidence of a mathematical mistake.

Retractions maintained: A previous apparent missing overbar in the prism’s $C_6 \vee \overline{K_2}$ notation was a text-extraction/reading mistake; the PDF was checked and that charge withdrawn. A previous allegation of absent exact quarter-flux holonomy evidence was narrowed after the integer certificate was examined. Neither is counted as an error here.

4.3 Optional exposition improvements

1. Display $G = \text{diag}(e^{i\chi_j})$ and $\Delta_{\theta'} = G\Delta_{\theta}G^*$ with $\theta'_{jk} = \theta_{jk} + \chi_j - \chi_k$ explicitly; define A_{hs} .
2. Add a short roadmap before the quarter-flux proof: gauge, Gram identity, block reduction, intertwiner, magnetic rotations.
3. Give one explanatory sentence about spherical integral cycle generation, and optionally include the new fourth-moment identity as a cross-check rather than a claimed central theorem.

5 Special Protocol Results

Gauge protocol (M15)	Status		Evidence
$U(1)$ connection, oriented face holonomies	Pass		Sec. 6; T09
Spherical product constraint and class existence	Reviewed proof		Prop. 6.1
Vertex-gauge transformation displayed	Partial		Suggested addition
Quarter-flux Gram identity	Exact check	finite	T06
Alternative gauge, covariance and charge symmetry	Pass		T05, T09
Improper rotation qualification	Pass (prior)		Sec. 6.1
Algebra protocol (M18)	Status		Evidence
Ordinary characteristic polynomials	Exact check	finite	T02
Special-flux characteristic polynomials	Exact check	finite	T08
Finite-group magnetic rotation representation	Exact (prior)	check	24 rotations/576 products
Length-three/length-four closed walks	Exact check	finite	T04
Lean or other proof-assistant validation	Not performed		Disclosure R6

Proof status. “Exact finite check” means a replayable exact symbolic/integer computation of the stated finite object, *not* a Lean-kernel theorem proof. There is no recorded checker build, trust-policy inspection, theorem mapping, or `#print` axioms output. We therefore attach *no* machine-validated label to any theorem. A proof-checker omission is not a counterexample.

Mathematical claim / location	Evidence considered	Machine-proof status
Prop. 2.1 graph counts	Geometric derivation; T01	Not machine checked
Thm. 3.1 spectrum	Orbit proof; T02, T03	Not machine checked
Lemma 3.2 representation decomposition	Human group-action argument; prior checks	Not machine checked
Prop. 4.1 weights, Thm. 5.1 K	Hand proof; inherited numeric/exact checks	Not machine checked
Prop. 6.1 spherical gauge classification	Tree/cycle-space proof	Not machine checked
Prop. 6.2 thirteen spectra	Exact third moment; T04, T05	Not machine checked
Thm. 6.3 special flux spectra	Gram block proof; T07, T08	Not machine checked
Prop. 6.4 Gram identity	Symbolic exact T06	Not machine checked
Magnetic rotations/intertwiner	Prior exact rotation checks	Not machine checked
Sec. 7 Fedorov table	Prior exact verifier	Not machine checked

6 Optimization Process Summary

What evolved. We actually executed a bounded *review-search* controller: 13 persistent logical agents numbered 0–12 in the original ternary tree; each retained Current, Pocket and a behavioural method policy. A generation-start snapshot provided tree-adjacent donor Pockets; odd generations used Current and even generations Pocket as the local parent. Recombination inherited supported task results and method priorities, while task-specific local search executed new checks in subprocesses. Selection and upward propagation used *strict Pareto dominance only*. All findings

(including adverse ones) entered a shared ledger regardless of Pocket retention. There were no synthetic verification passes.

What did not evolve. We did not instantiate 13 independently reasoning LLMs, vary the mathematical truth by voting, calibrate ReviewBot against human referees, perform Lean proof checking, or evaluate competing memetic policies on held-out papers. The operator set is a concrete pilot adaptation, not a claim to have recovered every detail of a historical implementation.

The pilot’s process vector was $(C, V, D, -U)$: C is the sum of frozen task priorities (3 for central checks, 2 for supporting) covered by the candidate; V counts completed challenge tasks, D is the count of distinct checking methods represented, and U counts unresolved *review contradictions* (zero in this finite task experiment). It is an exploratory process proxy, not a calibrated predictor of truth. The common frozen task universe has total coverage weight 36.

Gen.	Fresh tasks	Pocket updates	Tree promotions	Root $C/36$	New tasks (identifiers)
0	0	0	0	0/36	Initialization
1	3	3	3	2/36	T10, T01, T13
2	3	6	4	5/36	T02, T04, T12
3	3	5	4	7/36	T03, T07, T14
4	3	6	4	16/36	T05, T09, T06
5	2	10	1	17/36	T11, T08
Total	14	30	16		

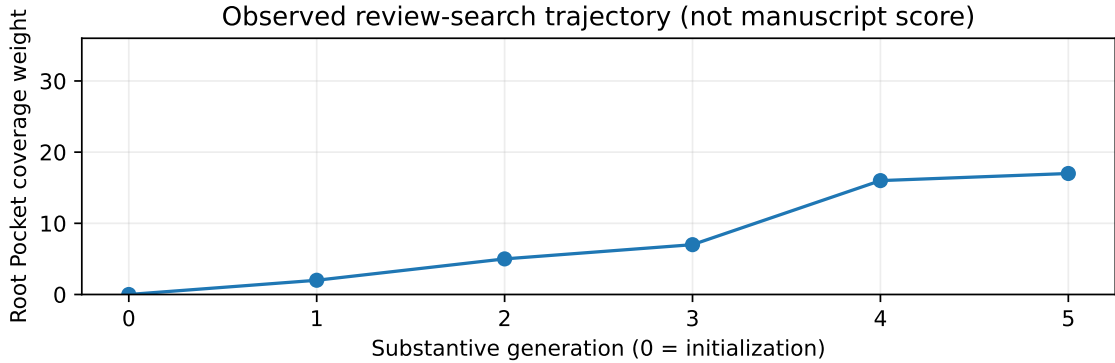


Figure 2. Actual root Pocket coverage progression, not a manuscript-quality trajectory or convergence simulation.

The root Pocket retained seven of the fourteen freshly verified task IDs, because strict dominance does not assemble incomparable child plans into a fictitious best single Pocket. The root’s *report* nevertheless incorporates all fourteen independently preserved ledger results. The run stopped after generation 5 because *all fourteen frozen tasks had been executed*; the eight-generation upper budget was not reached and no three-generation stability convergence was observed. A new extended review task universe would be necessary before performing additional substantive generations. Task execution outcomes and process trace are available in the archive.

7 Overall Recommendation

Accept with minor revisions for an appropriate specialist mathematics or mathematical-physics outlet, contingent on correction and confirmation of R1–R6. This recommendation is narrower than an assertion that the manuscript has been formally proven or that its UFFT motivation has physical support.

The technically impressive part is not simply a list of eigenvalues. It is the sequence of explanations: finite face incidence $\rightarrow$ orbit blocks $\rightarrow$ eigenspace structure $\rightarrow$ magnetic gauge $\rightarrow$ exact Gram reduction $\rightarrow$ intertwining and rotations $\rightarrow$ thirteen-spectrum classification. Both the previous adapted review and the new finite calculations support that story.

The six corrections are repairable without changing the principal spectral formulas. A follow-up review should focus on the corrected release, fresh installation in a clean environment, and claim-to-checker proof coverage, not on expanding ReviewBot architecture at the expense of reading Draft 5. Historical novelty and broad physical conclusions are outside the evidence.

Reproducibility and review provenance

The accompanying ZIP supplies the manuscript, this report source, actual check programs, first-pass prior review reports, complete CSV/JSON Current–Pocket/PocketPropagation trace, all fourteen fresh check results, two plotted diagnostics and a manifest with file hashes. The `run_memetic_review.py` entrypoint is the exact Python procedure used here. Its controller carries out real data/check operations but does not implement a service with 13 independent human-like reviewers. All new facts are marked as newly verified; prior third-party/agent verification logs remain separately attributed.